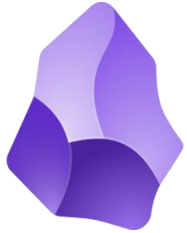


A short introduction to Obsidian, Overleaf and Zotero

Introducing the programs



Obsidian: Notes app able to create mind maps, link notes together and allows the addition of citations to your notes



Zotero: App to collect, organize, annotate and cite papers from



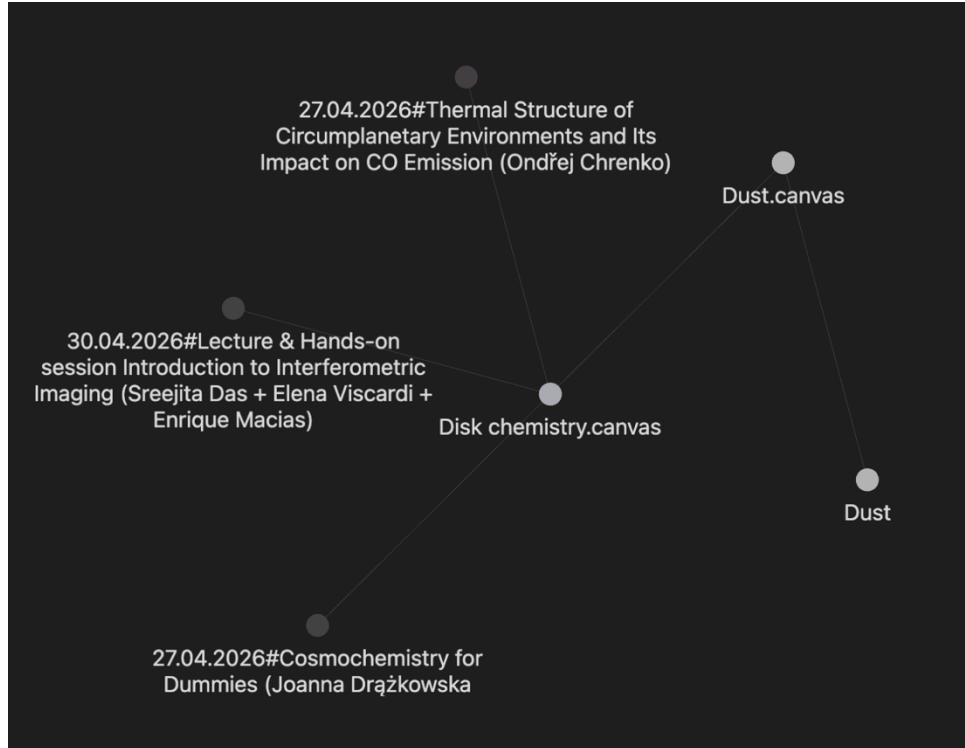
Overleaf: Cloud-based latex editor

Connect them



Clickable links!

Introduction to Obsidian

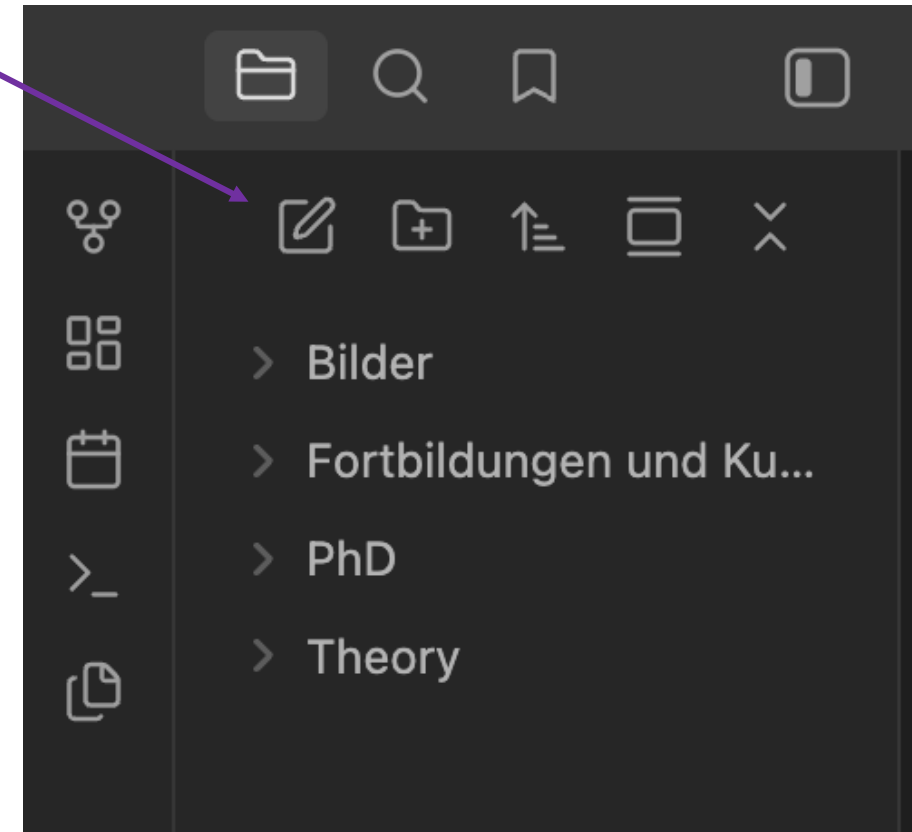


New Note

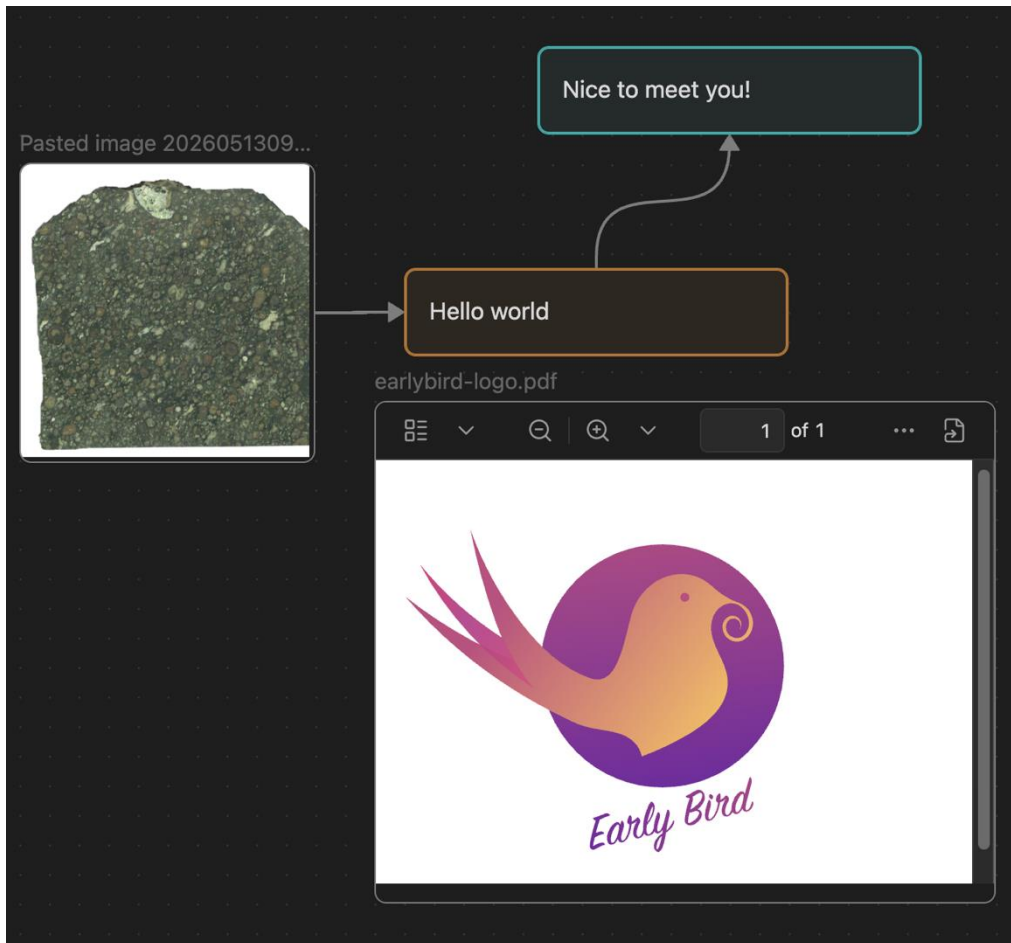
Graph view

File tree

Hide Sidebar



Introduction to Obsidian



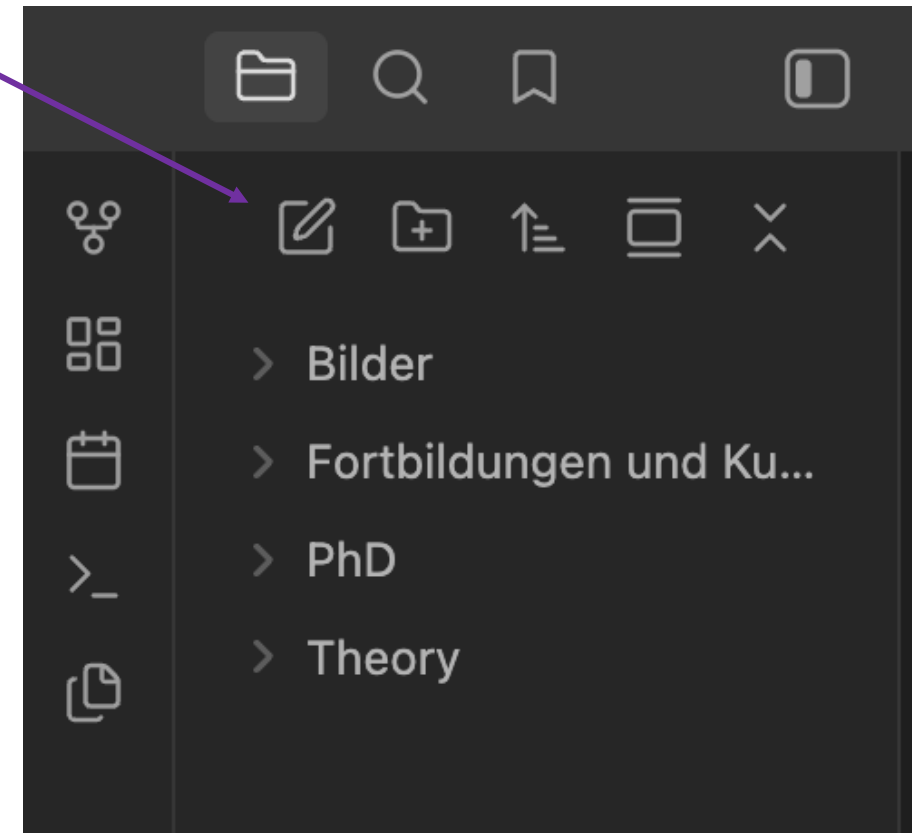
New Note

Graph view

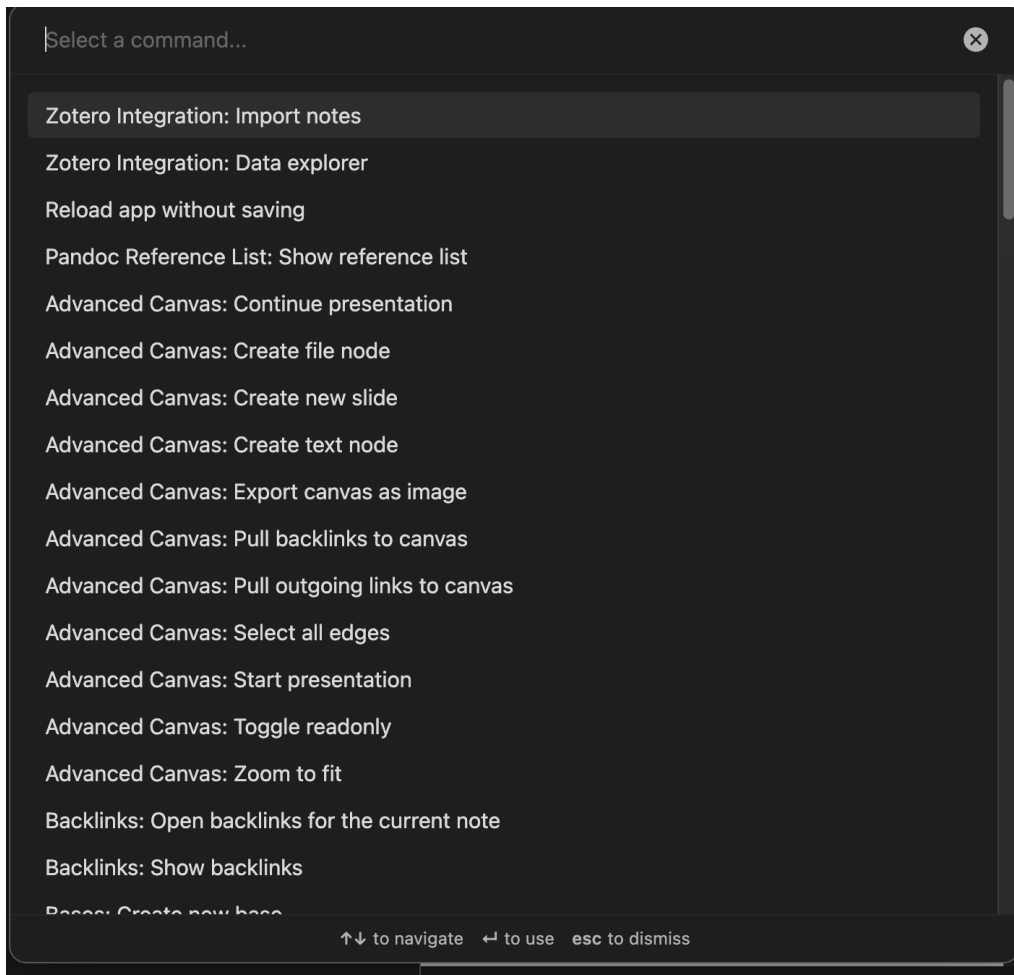
New canvas/
mind maps

File tree

Hide
Sidebar



Introduction to Obsidian



New Note

Graph view

New canvas/
mind maps

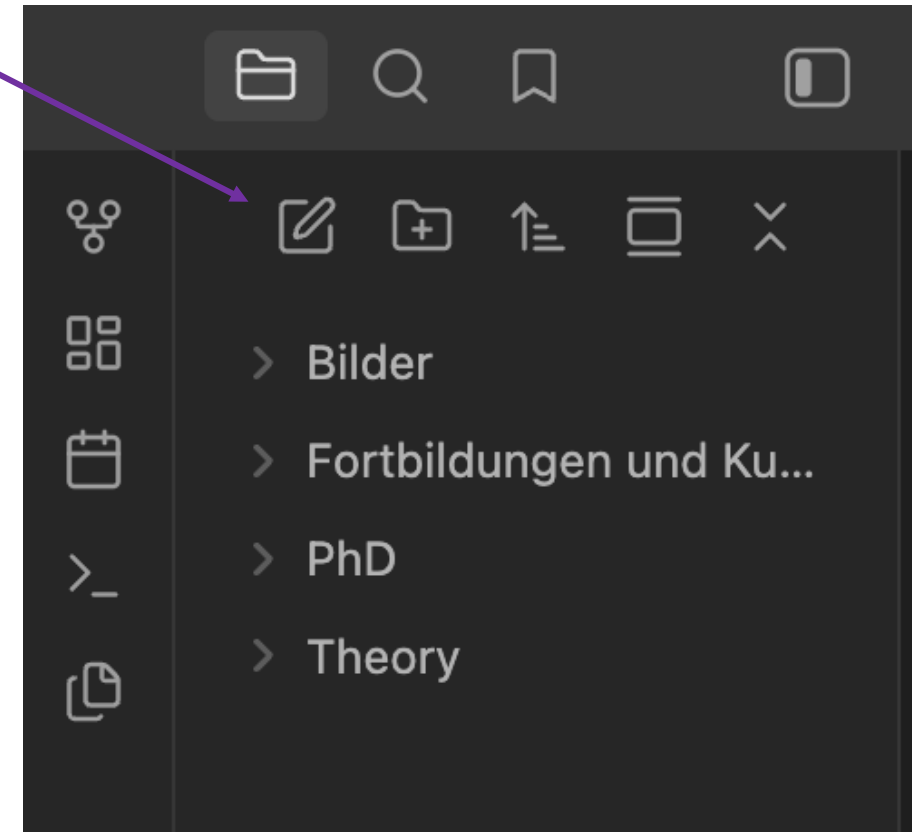
Daily note

Command
palette

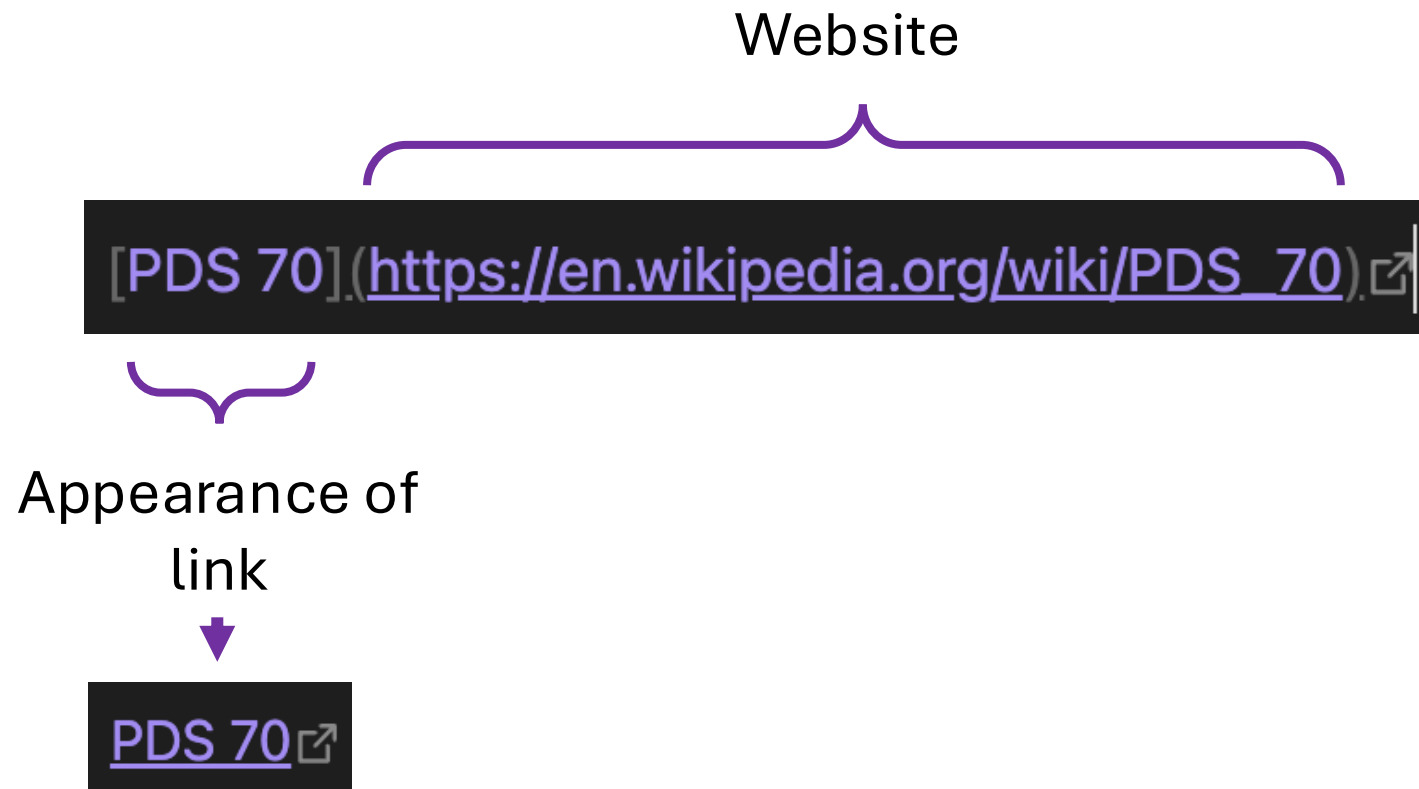
Templates

File tree

Hide
Sidebar



Feature: Linking websites



Feature: Connecting notes/canvases

Chapter in the note with “#” in front

[[27.04.2026#Planet Formation Initiated by Infall-Induced Pressure Bumps (Haichen Zhao)|Haichen talk]]

Name of the
Note

Appearance of
link

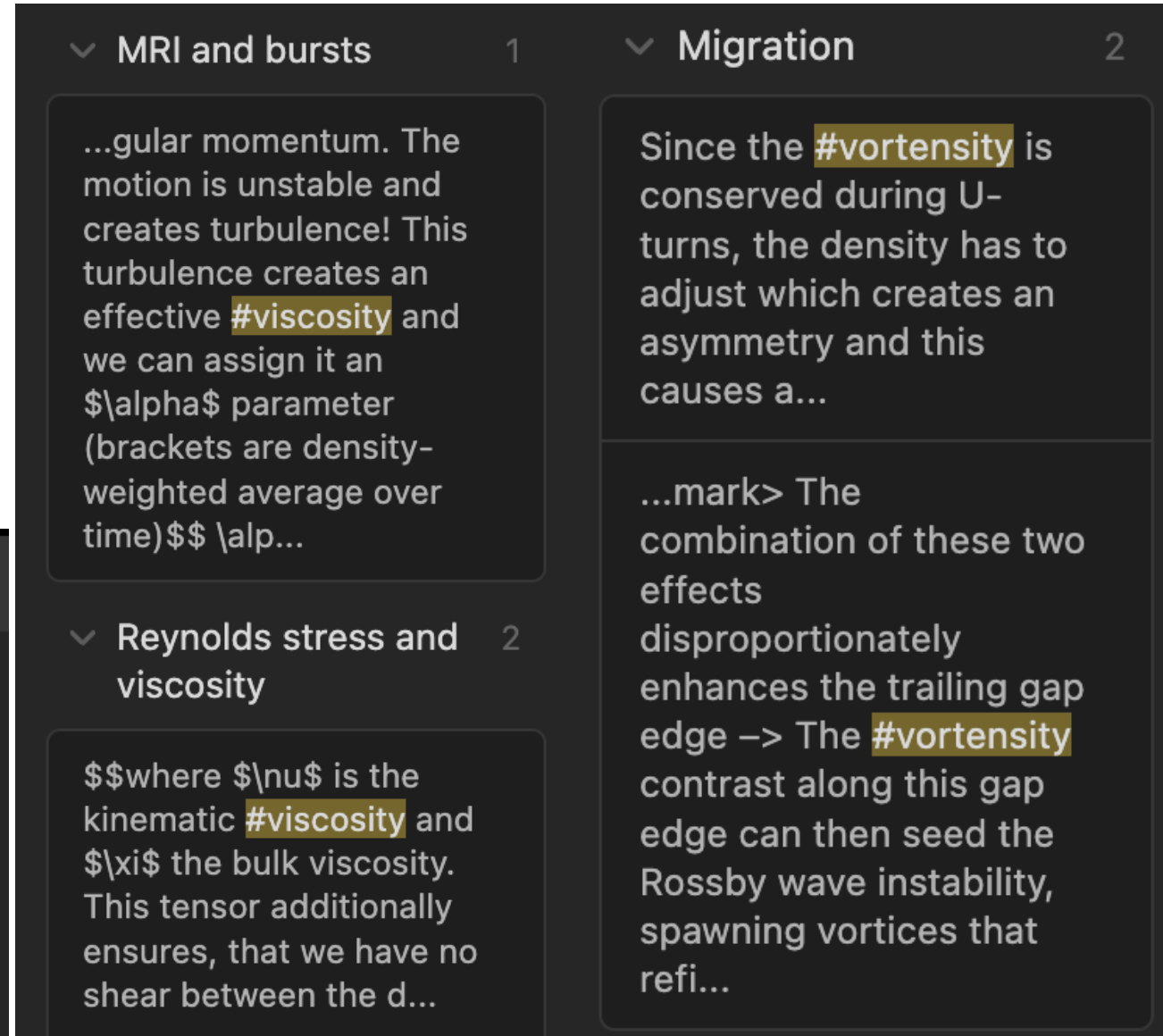
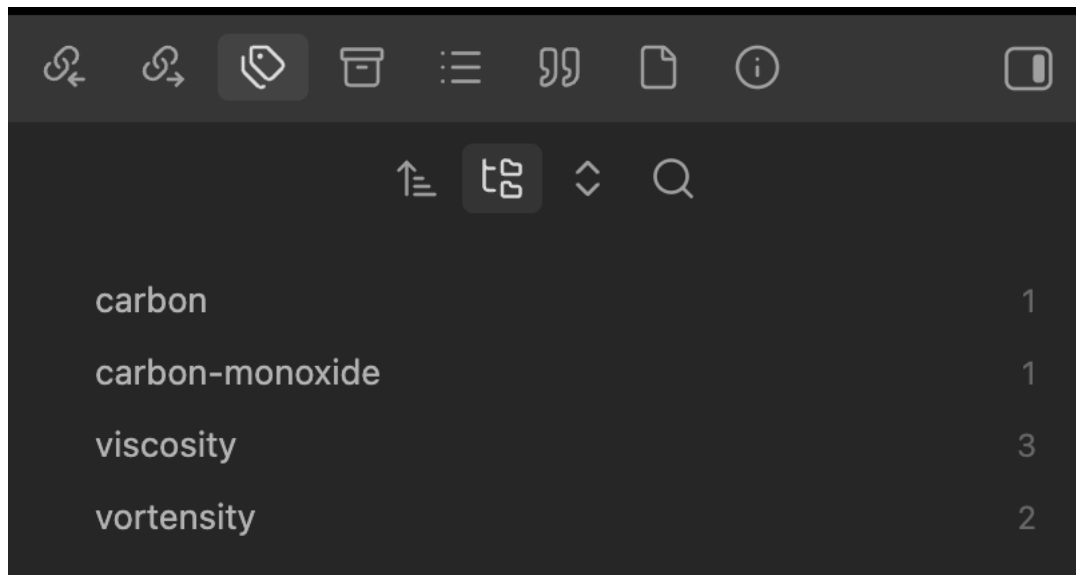
Haichen talk

→ Connecting notes creates a link in the graph view

Feature: Tags

Place Tags in your text with #

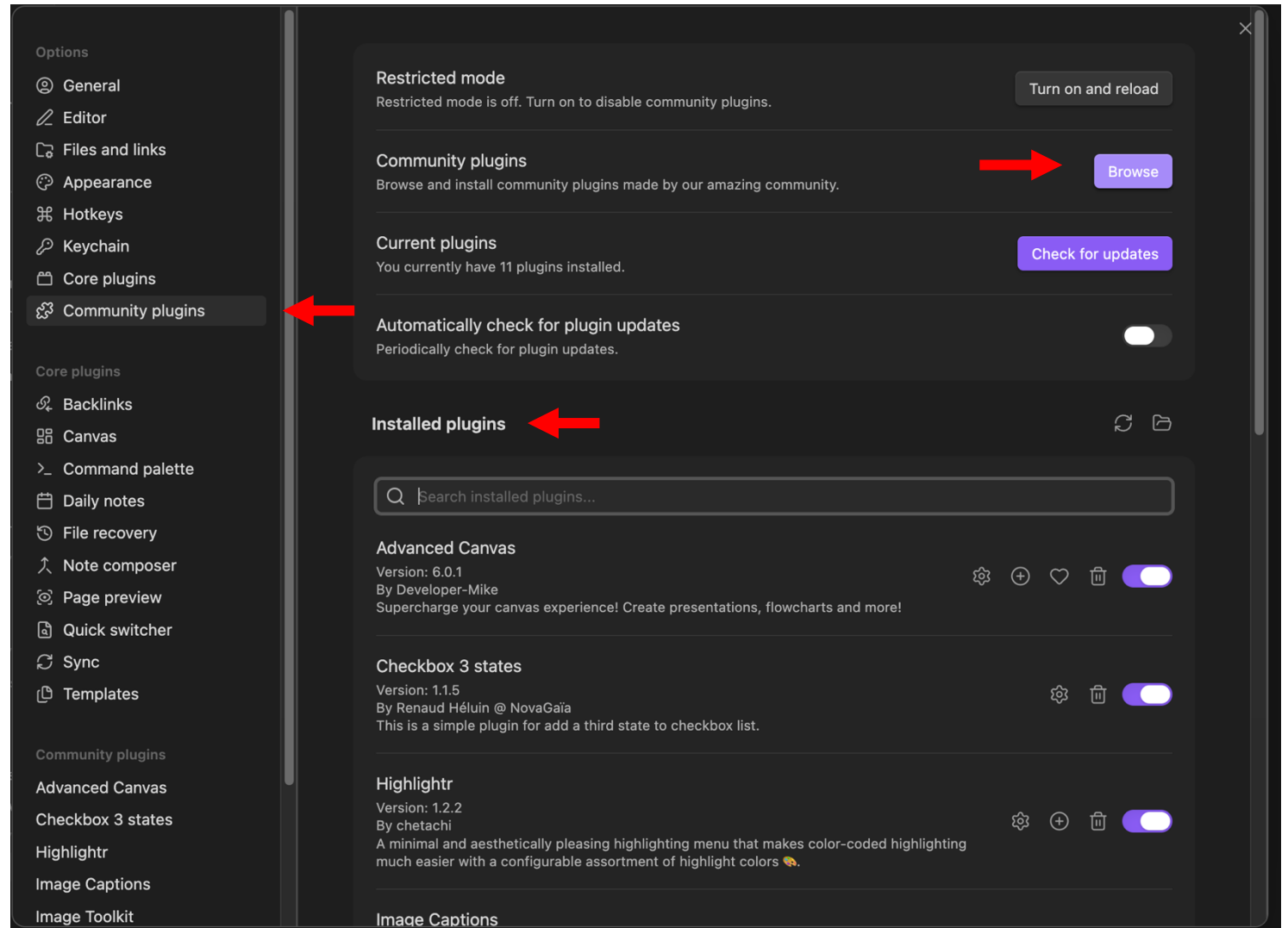
→ Tag list shows all your tag and where you placed them



Community Plugins in Obsidian

1. Open settings
2. Click “Community plugins”
3. Click “Browse”

List of installed plugins below



Community Plugins in Obsidian: Basics

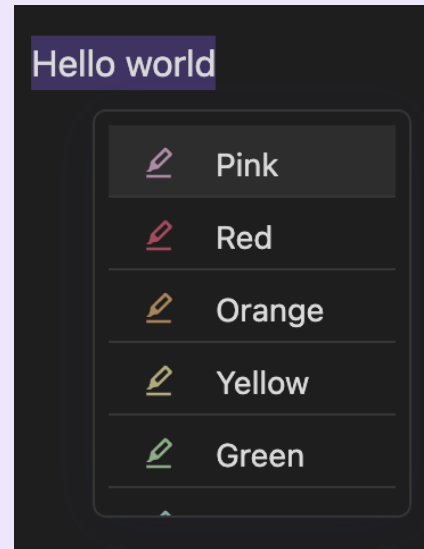
Language Tool

Corrects spelling mistakes

→ Better than apple spelling check

Highlightr

More highlighter colours



Symbol Prettifier

Text	Symbol
->	→
<-	←
<->	↔
<=>	↔
<=	≤
=>	⇒
=<	≤
>=	≥
+ -	±
- +	±

Community Plugins in Obsidian: Basics

cMenu

Adds mini text
formatting toolbar

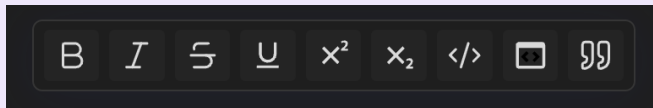


Image Toolkit

Allows us to zoom into
Images and has a dark
mode option

Advanced Canvas

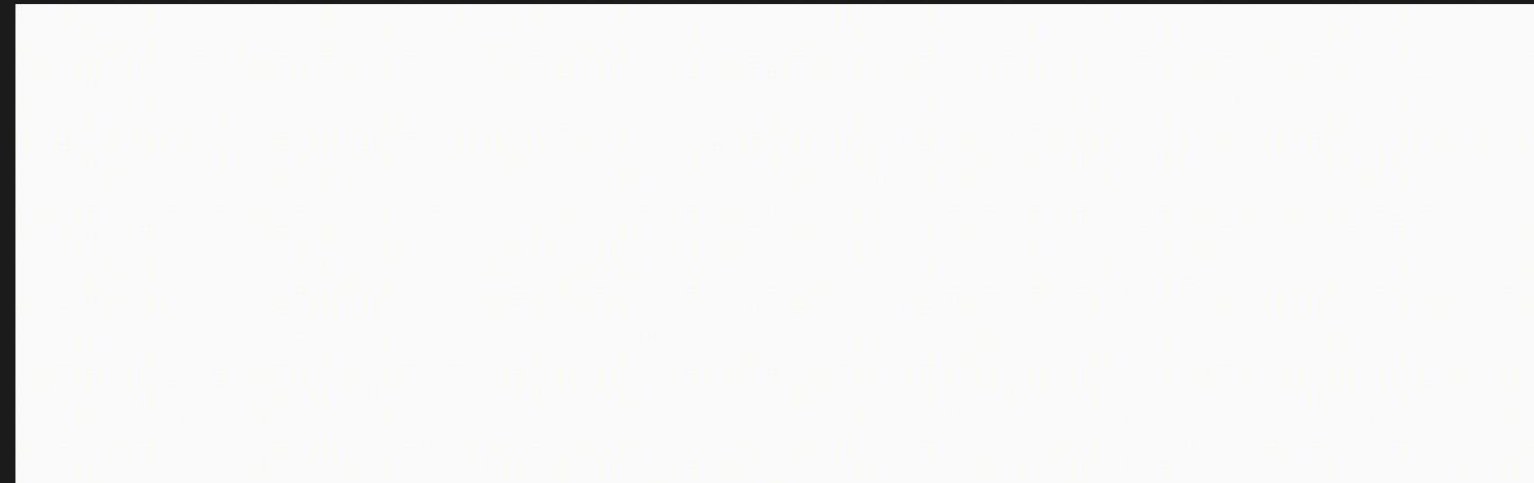
Adds new cosmetics
and commands for
canvases

Community Plugins in Obsidian: Latex Suite

Fast typing in Latex

A plugin for Obsidian that aims to make typesetting LaTeX math as fast as handwriting.

Inspired by [Gilles Castel's setup using UltiSnips](#).



The screenshot displays a LaTeX document with a complex mathematical proof. The text is rendered in a serif font, with mathematical symbols and formulas clearly visible. The document appears to be a formal proof, possibly related to number theory or algebra, given the use of terms like "prime", "divides", and "coprime". The proof involves several lines of text and mathematical expressions, including a large fraction and a summation. The overall layout is clean and professional, typical of a mathematical manuscript.

Introducing Zotero

Store, sort and
annotate all your
favourite papers

The screenshot shows a web browser window with the URL `ui.adsabs.harvard.edu/abs/2024MNRAS.534.1394C/abstract`. The browser's address bar and tabs are visible at the top. A Zotero Connector extension popup is open in the top right corner, displaying the text "Zotero Connector" and "Kann immer Daten auf dieser Website lesen und ändern". The main content area shows the abstract of a paper titled "Early stages of gap opening by planets in protoplanetary discs" by Cordwell, Amelia J. and Rafikov, Roman R. The abstract text discusses the formation of gaps in protoplanetary discs. On the right side, there are sections for "FULL TEXT SOURCES" (OUP, Preprint), "DATA PRODUCTS" (SIMBAD), and "RELATED MATERIALS" (Erratum). At the bottom right, there is a "GRAPHICS" section with four subplots (a, b, c, d) showing the evolution of the gap profile over time. A "View page in SciX" button is visible below the graphics.

VIEW

- Abstract
- Citations (11)
- References (56)
- Co-Reads
- Similar Papers
- Volume Content
- Graphics
- Metrics
- Export Citation

QUICK FIELD: Author First Author Abstract Year Fulltext All Search Terms

author:"Cordwell" author:"Rafikov"

Back to results

Early stages of gap opening by planets in protoplanetary discs

Show affiliations

Cordwell, Amelia J. ; Rafikov, Roman R.

Annular substructures in protoplanetary discs, ubiquitous in sub-mm observations, can be caused by gravitational coupling between a disc and its embedded planets. Planetary density waves inject angular momentum into the disc leading to gap opening only after travelling some distance and steepening into shocks (in the absence of linear damping); no angular momentum is deposited in the planetary co-orbital region, where the wave has not shocked yet. Despite that, simulations show mass evacuation from the co-orbital region even in inviscid discs, leading to smooth double-trough gap profiles. Here, we consider the early time-dependent stages of planetary gap opening in inviscid discs. We find that an often-overlooked contribution to the angular momentum balance caused by the time-variability of the specific angular momentum of the disc fluid (caused, in turn, by the time-variability of the radial pressure support) plays a key role in gap opening. Focusing on the regime of shallow gaps with depths of $\lesssim 20$ per cent, we demonstrate analytically that early gap opening is a self-similar process, with the amplitude of the planet-driven perturbation growing linearly in time and the radial gap profile that can be computed semi-analytically. We show that mass indeed gets evacuated from the co-orbital region even in inviscid discs. This evolution pattern holds even in viscous discs over a limited period of time. These results are found to be in excellent agreement with 2D numerical simulations. Our simple gap evolution solutions can be used in studies of dust dynamics near planets and for interpreting protoplanetary disc observations.

Publication: Monthly Notices of the Royal Astronomical Society, Volume 534, Issue

FULL TEXT SOURCES

- OUP
- Preprint

DATA PRODUCTS

- SIMBAD (1)

RELATED MATERIALS (1)

- Erratum: 2024MNRAS.535.1695C

GRAPHICS

Simulation

View page in SciX

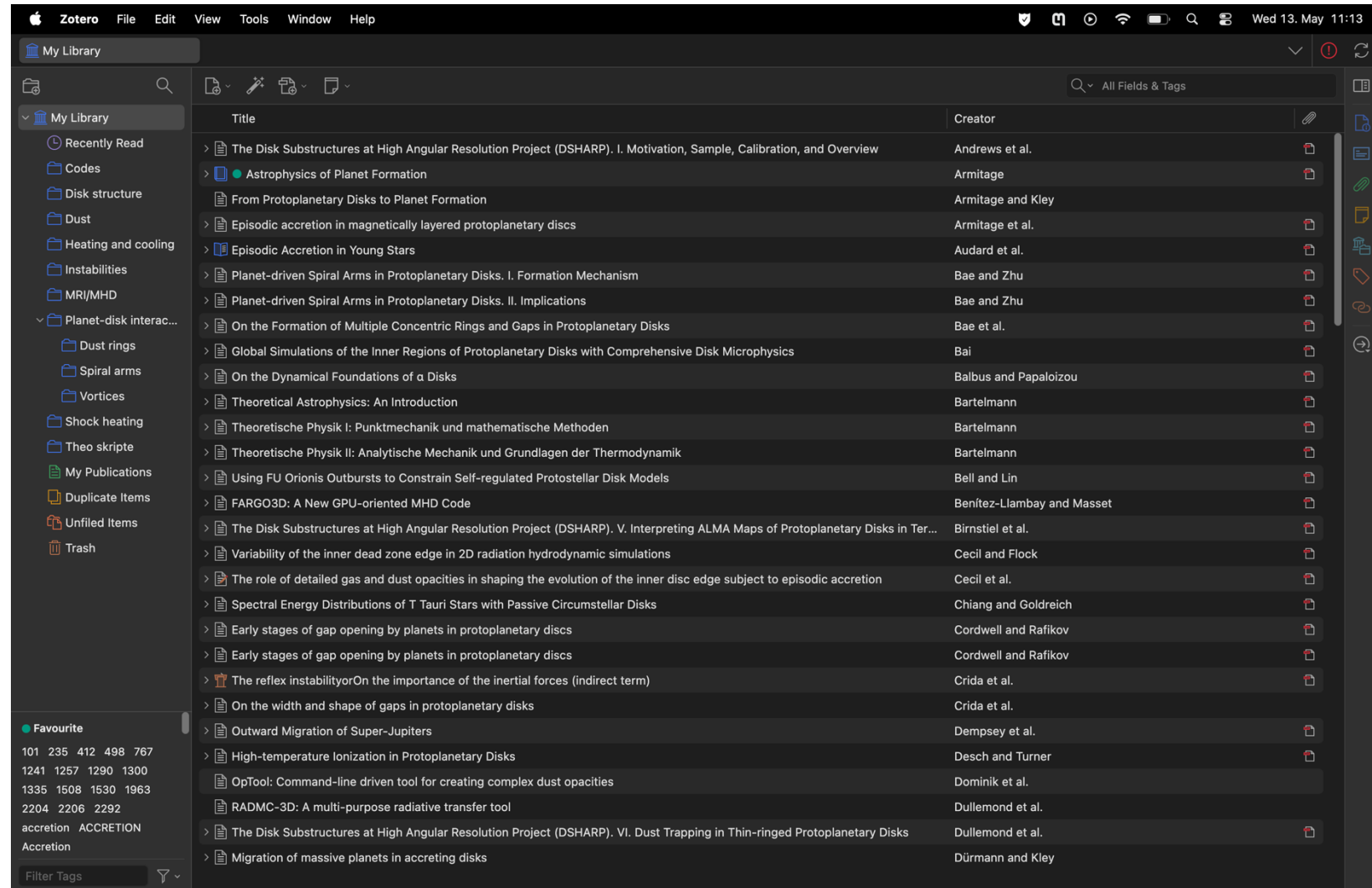
Click to view more

Introducing Zotero

Store, sort and
annotate all your
favourite papers

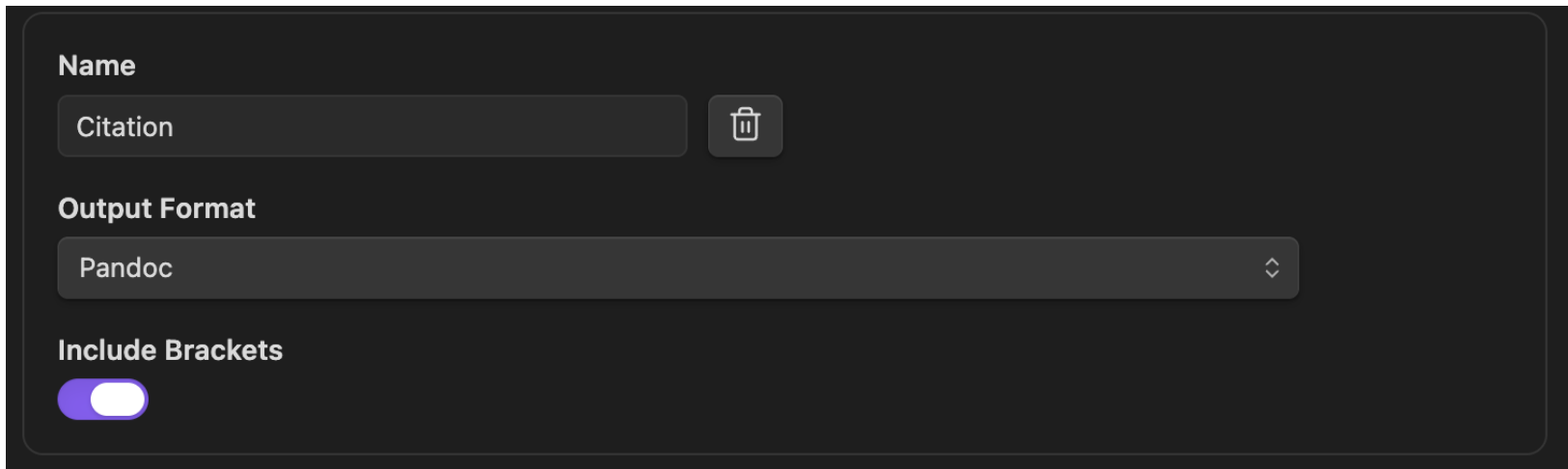
→ Firefox Plugin

→ Community
plugins



Connecting Obsidian and Zotero

1. Install "Zotero Integration" in Obsidian
 - Download PDF utility → allows PDFs to be read
 - Set Database to Zotero & define a citation format



The screenshot shows the 'Zotero Integration' settings panel in Obsidian. It has a dark theme. The 'Name' field contains 'Citation' and has a trash icon to its right. The 'Output Format' is a dropdown menu currently showing 'Pandoc'. The 'Include Brackets' toggle switch is turned on, indicated by a blue bar.

Name

Citation 

Output Format

Pandoc 

Include Brackets

☒

Connecting Obsidian and Zotero

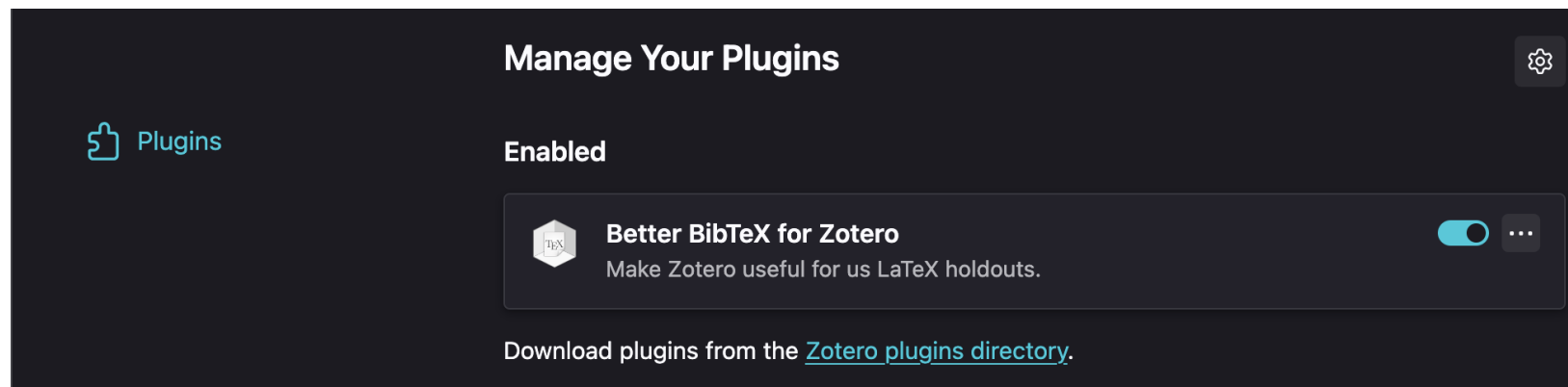
2. Install “Pandoc reference list” and “Pandoc Plugin” in Obsidian

- Install Pandoc additionally over your terminal
- Link to Pandoc to your Zotero Database
- Chose a citation style (I prefer “Cite them right 10th edition – Harvard)

Connecting Obsidian and Zotero

3. Install “Betterbibtex” in Zotero

- Go to “Tools”
- Go To “Plugins” and select “Betterbibtex”



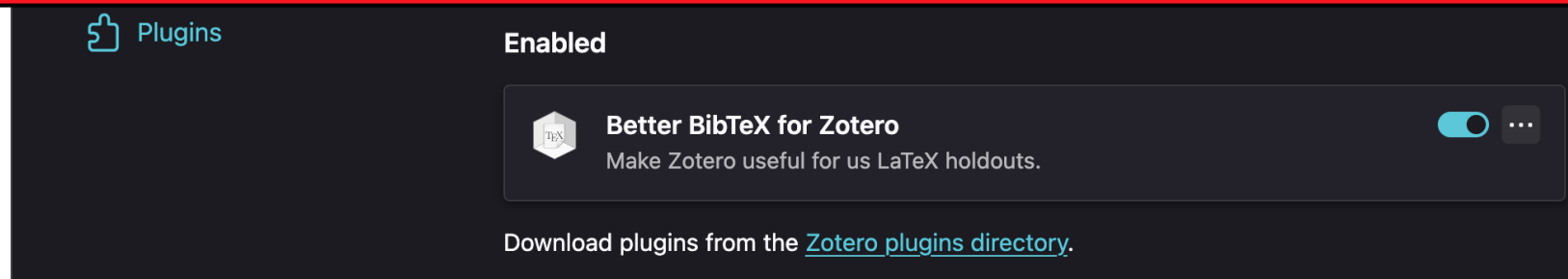
Connecting Obsidian and Zotero

3. Install “Betterbibtex” in Zotero

Pitfall:

Problems connecting and updating your Zotero Library

→ Better export your library regularly and let obsidian pull the references from a file

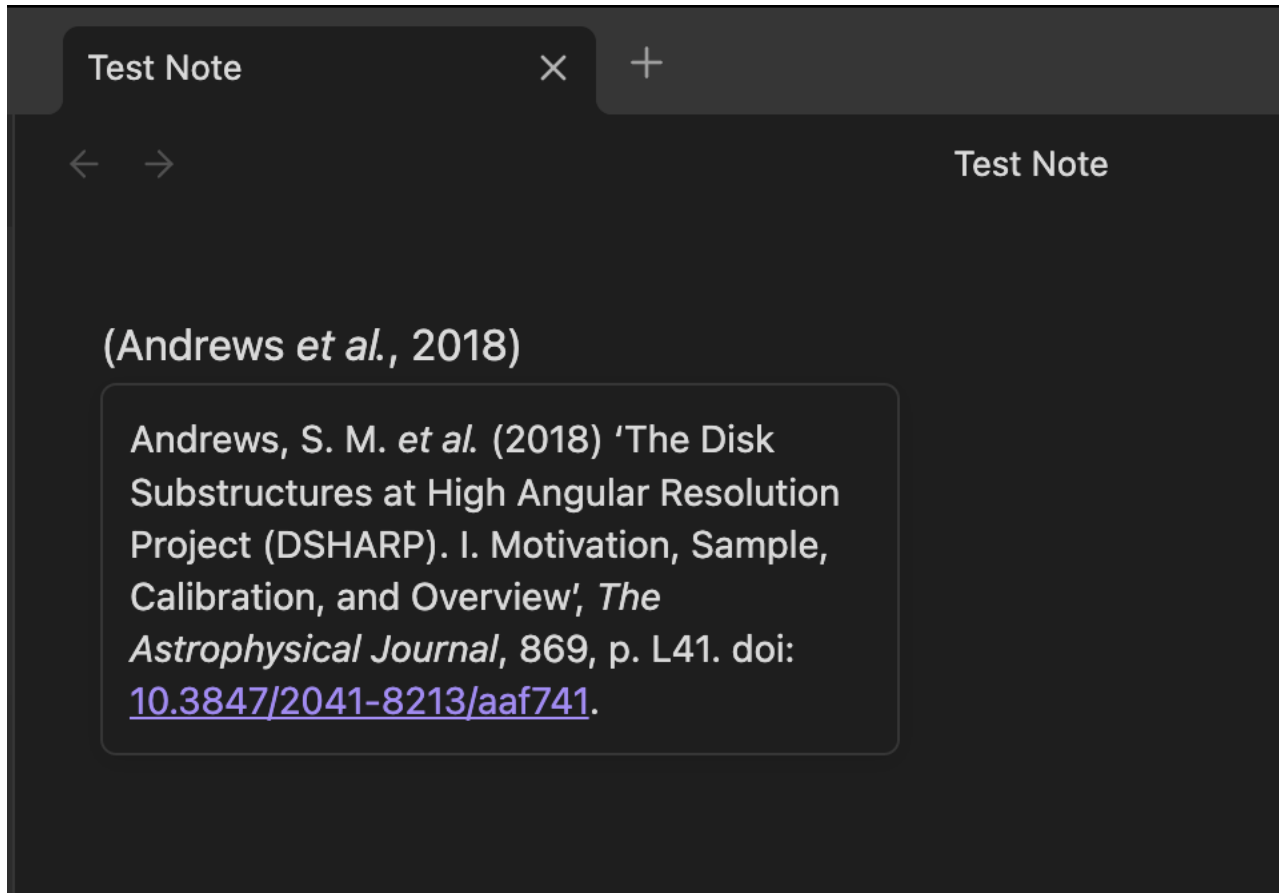


Connecting Obsidian and Zotero

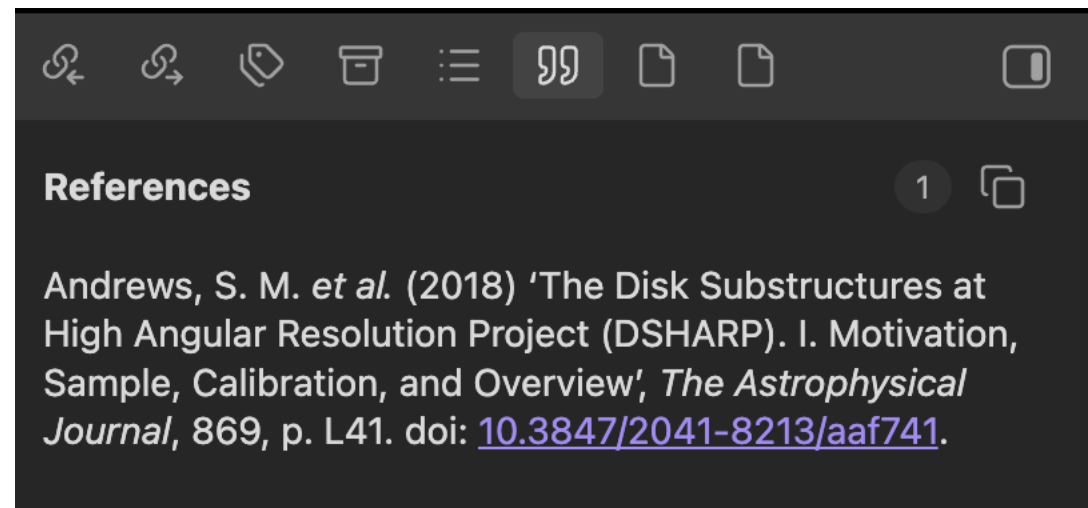
Usage:

1. Click on Command Pallett
 2. Click on/search for “Zotero Integration: {Your Citation name}”
 3. Select Paper from Zotero
- Link can be found on the right under references

Connecting Obsidian and Zotero



Show reference tab



Introducing Overleaf

Cloud-based latex editor

- Saves your Documents even if your PC crashes
- Allows for easy collaboration
- Access to a lot of [templates](#)
- Decent AI feature (but behind a paywall)

Connecting Overleaf and Zotero

1. Change email address to LMU email address for overleaf in “account setting”
→ enables overleaf premium
2. Open your project
3. On the left side, click in “integration” 
4. Click on Zotero and log into your account
5. Add new file: from Zotero → imports a .bib file with all your references

Unlink when working on a paper, so others can add papers to the library

